

Do LLM Judges Prefer LLM-Generated Plans? Order Effects and Criterion Decomposition in Pairwise LLM Evaluation

Cameron Caputa

Abstract

LLMs are increasingly used as judges for open-ended NLP artifacts, but pairwise judgments may reflect position, source, or presentation cues rather than substantive quality. We test whether LLM judges prefer directly LLM-generated trail-running plans over programmatically structured controls when both are structurally matched and rendered through the same judge-facing schema. Across 250 matched pairs, four Qwen/Gemma judges, five runs, and AB/BA order swaps, LLM-source plans win only 36.53% of 10,000 forced judgments. The dominant effect is positional: LLM-source plans win 57.54% as Plan A but only 15.52% as Plan B. Marker-level evaluation also favors programmatic controls on every criterion, with near parity only on training specificity. Structural matching therefore improves source-preference studies but does not make pairwise LLM judgments reliable; order swaps, criterion decomposition, and checks for presentation leakage should be core design requirements.

Code is available at: <https://github.com/MauriceCC04/llm-judge-self-preference>

1 Introduction

LLM-as-judge evaluation is widely used for open-ended NLP outputs because many artifacts lack a single reference answer. However, pairwise LLM judging can reflect position bias, verbosity bias, self-enhancement bias, or source cues rather than substantive quality (Zheng et al., 2023; Gu et al., 2024). This paper studies a narrower source-preference question: when artifacts are structurally matched and source-masked, do LLM judges prefer outputs produced by direct LLM-source generation over outputs produced by a programmatic structural generator?

We test this question in trail-running training-plan generation. Trail-running plans provide a use-

ful testbed because they combine objectively describable structure—weekly load, rest days, hard-day spacing, and long-run placement—with subjective presentation qualities such as clarity, specificity, and rationale quality. The goal is not to determine which generator produces objectively better coaching plans. Instead, we ask whether a common LLM-as-judge setup can support a source-preference claim once structure, order, and presentation cues are controlled. Because the programmatic arm may include LLM-written narrative fields, the comparison is direct LLM-source generation versus programmatic structural generation, not fluent LLM prose versus empty text.

This study makes three contributions. First, it provides a controlled source-preference test using 250 structurally matched LLM-vs-programmatic pairs and 10,000 order-balanced pairwise judgments. Second, it shows that the main threat to interpretation is order sensitivity: LLM-source plans win 57.54% as Plan A but only 15.52% as Plan B. Third, it decomposes the result by athlete band and explicit evaluation markers, showing that the aggregate programmatic advantage is strongest for beginner profiles and that programmatic controls win every marker. Together, the results suggest that structural matching can make source-preference comparisons more controlled, but it cannot by itself make pairwise LLM judgments stable or reliable.

2 Related Work

LLM-as-judge evaluation uses models to score or compare open-ended artifacts when reference answers are unavailable or incomplete. MT-Bench and Chatbot Arena show that pairwise LLM judging can scale evaluation, but they also identify position bias, verbosity bias, self-enhancement bias, and reasoning limitations (Zheng et al., 2023). Surveys of LLM-as-judge systems similarly emphasize prompt sensitivity, source cues, position effects,

and aggregation choices (Gu et al., 2024). These studies motivate treating the judge itself as an experimental object rather than as a neutral measuring device.

Reliability work argues that repeated judgments are needed because single LLM judgments may not be stable across samples or settings (Schroeder and Wood-Doughty, 2025). This is especially important for pairwise judging: AB/BA swaps can balance the aggregate source estimate while individual decisions remain unstable. We therefore distinguish an order-balanced aggregate preference from pair-level reliability across swapped presentations.

Criteria-decomposed evaluation methods such as HD-Eval motivate breaking holistic preferences into explicit dimensions rather than treating a single overall judgment as self-explanatory (Liu et al., 2024). Related work on learner-created artifacts and creative writing similarly shows that LLM evaluation quality can vary across rubric dimensions, domains, and prompt formats (Tian et al., 2024; Kim and Oh, 2025). Our marker pass adopts this diagnostic motivation for training plans: it tests whether the source effect appears in structure-sensitive criteria, such as recovery safety and load progression, or in presentation-sensitive criteria, such as clarity and explanation quality.

The gap for this paper is narrower than general LLM-as-judge benchmarking. Prior work establishes that LLM judges can be biased or unstable; we ask whether a source-preference claim still holds when the compared artifacts are matched on visible structure, presented through the same schema, evaluated in both AB and BA orders, and decomposed into explicit criteria. Crowd-comparison work instead tries to improve judges with richer anchors (Zhang et al., 2025); our goal is complementary, auditing a standard pairwise setup rather than proposing a new judge architecture.

3 Experiments

The final candidate pool contains 192 Qwen-source plans, 192 Gemma-source plans, and 640 programmatic plans. The programmatic arm samples structural constraints such as hard-day count, rest-day placement, recovery capacity, race phase, and load. A *fixture* is one experimental cell: an athlete band crossed with readiness, recovery capability, and race phase. Athlete band is treated as a substantive matching variable because it changes both the generation context and what counts as an appropriate

weekly structure. An early generation audit found substantial within-cell duplication, so the conditioning space was expanded with athlete profiles from beginner/low-volume to high-volume competitive runners.

Each LLM-source plan is matched to one programmatic plan using a source-neutral structural score over plan length, total minutes, rest days, hard days, quality days, long runs, duration plausibility, load-band fit, and hard-day spacing. The structural score is a matching aid, not a quality label: it measures visible weekly-plan similarity under fixture-conditioned expectations. Candidate pairs must share fixture ID, athlete band, readiness, recovery capability, race phase, plan length, and style, and must fall within a structural-score tolerance of 2.0. The matcher then selects a target-cardinality minimum-cost matching of 250 pairs using weighted structural feature distances and soft caliper penalties, where a soft caliper penalizes large feature gaps without forbidding them. Hard calipers were not enabled in the frozen primary match because they would have reduced coverage below the target. The final matched set has mean structural-score gap 0.3076 and maximum gap 2.0.

Both plans are rendered through the same canonical masked judge-facing schema, which removes source/provenance metadata and preserves day-level structure plus short workout and purpose summaries. The frozen evaluation manifest stores source family, plan IDs, order, judge, run, and structural-score metadata outside the prompt, while the judge sees only neutral `plan_a` and `plan_b`. This is structural-score matching rather than full quality equalization: unmeasured coaching logic, wording style, and residual presentation cues may still differ between sources.

Four local judges, `qwen_7b_judge`, `qwen_14b_judge`, `gemma_4b_judge`, and `gemma_12b_judge`, evaluate each matched pair across five runs and two AB/BA orders, producing 10,000 forced judgments. The primary outcome is whether the judge selects the LLM-source plan. A second pass evaluates nine *markers*, or criterion-level preferences—plan coherence, training specificity, load progression, recovery safety, quality session design, endurance development, readiness alignment, clarity/actionability, and explanation quality—allowing ties and producing 90,000 marker decisions; the full marker prompt is reproduced in Appendix B. Marker outputs were schema-validated; rare

Finding	Evidence
No aggregate LLM-source preference	LLM-source wins 36.53%; programmatic wins 63.47%.
Athlete band moderates the effect	A1: 12.50%; A4: 53.32%.
Order dominates interpretation	57.54% as Plan A vs. 15.52% as Plan B.
Markers favor programmatic controls	Best LLM marker is specificity at 48.54%; explanation quality is 32.21%.
Leakage filtering is stable	Clean subset LLM win rate is 34.44%.

Table 1: Compact summary of the main results. Detailed tables are in Appendix G.

malformed categorical preferences were repaired from numeric scores and flagged. Because judgments repeat the same matched pairs across judges, runs, and orders, row-level win rates are descriptive rather than independent-sample inference. We therefore treat matched-pair behavior, order-swapped consistency, marker decomposition, athlete-band stratification, and leakage-filtered sensitivity as the main robustness checks. The leakage-filtered sensitivity analysis is a post hoc reanalysis that removes matched pairs flagged for presentation artifacts that could cue source despite masking; row-level confidence intervals appear only as descriptive scale checks in the appendix.

4 Results

Table 1 summarizes the main empirical findings. Overall, LLM-source plans do not win against structurally matched programmatic controls: judges select them in 36.53% of forced judgments, while programmatic controls win 63.47% (Appendix Table 4). This aggregate hides substantial athlete-band heterogeneity (Appendix Table 5). LLM-source plans win only 12.50% for A1 beginner fixtures and 24.55% for A2 developing fixtures, but reach 51.08% for A3 advanced fixtures and 53.32% for A4 competitive fixtures. Thus, the result is not a uniform programmatic advantage; it is concentrated in lower-volume athlete regimes where conservative structure may be more constrained.

The dominant methodological result is the order effect (Table 2; Appendix Table 6). LLM-source plans win 57.54% of judgments when shown as Plan A but only 15.52% when shown as Plan B, a 42.02 percentage-point swing. AB/BA balancing prevents a trivial confound between source and position, but it does not make the judge’s decision stable: many pair-level outcomes reverse when the same plans are swapped. Pair-run consistency shows the same problem (Table 3; Appendix Ta-

Judge	LLM as A	LLM as B	Swing
Gemma 12B	39.60%	19.60%	20.00 pp
Gemma 4B	62.00%	4.88%	57.12 pp
Qwen 14B	74.40%	8.40%	66.00 pp
Qwen 7B	54.16%	29.20%	24.96 pp
Overall	57.54%	15.52%	42.02 pp

Table 2: Order effect by judge. Rates are LLM-source win rates when the LLM-source plan appears as Plan A versus Plan B.

Diagnostic	Rate
Source-consistent programmatic winner	39.46%
Source-consistent LLM-source winner	12.52%
Position-consistent Plan A winner	45.02%
Training specificity, LLM win rate excl. ties	48.54%
Load/quality/endurance, LLM win rates excl. ties	35.11% / 33.44% / 32.81%
Explanation quality, LLM win rate excl. ties	32.21%

Table 3: Main stability and marker diagnostics. Full pair-run consistency and marker tables appear in Appendix Tables 7 and 8.

ble 7). Here, source-consistent means the same source wins in both swapped orders, while position-consistent means the same screen position wins after the swap. Across matched pair–judge–run units, programmatic plans are source-consistent winners in 39.46%, LLM-source plans are source-consistent winners in only 12.52%, and Plan A is the position-consistent winner in 45.02%. The experiment therefore estimates an order-balanced aggregate preference, not a stable pairwise decision mechanism.

Marker-level evaluation reinforces rather than reverses the aggregate result (Table 3; Appendix Table 8). Programmatic controls win every marker. The closest marker is training specificity, where LLM-source plans reach 48.54% of non-tied decisions, suggesting near parity on athlete- and trail-specific adaptation. Structure-sensitive markers favor programmatic controls more strongly: LLM-source plans win 35.11% on load progression, 33.44% on quality session design, and 32.81% on endurance development. Unexpectedly, explanation quality is lowest at 32.21%, despite the expected fluency advantage of LLM-source plans. This should not be read as evidence that LLMs are generally worse writers; rather, the shared schema compresses both sources into short workout and purpose summaries, and some controls include narrative fields, so this marker partly evaluates whether the rationale aligns with weekly structure rather than raw prose fluency. Tie behavior also matters: clarity/actionability and readiness alignment have many ties, while load progression and en-

duration development have fewer ties and clearer programmatic advantages. The marker pass therefore adds information beyond the forced-choice result: it shows that the programmatic advantage is not merely a single global preference, but is strongest on criteria tied to weekly training structure. A post hoc leakage-filtered sensitivity analysis is stable: LLM-source plans win 34.44% of clean-subset judgments, compared with 36.53% overall (Appendix Table 10).

5 Discussion and Limitations

The main result is methodological rather than domain-specific. Structural matching and shared schemas reduce obvious confounds, but they do not make pairwise LLM judgments stable. AB/BA balancing protects the aggregate estimate from source-position imbalance, yet individual decisions remain highly position-sensitive. Judge-level order swings range from 20.00 to 66.00 percentage points, so the effect is not reducible to one idiosyncratic judge. This matters for NLP evaluation settings where small numbers of pairwise LLM judgments are used for model selection, leaderboard ranking, or artifact evaluation. A source-preference study that reports only one order per pair could reach a very different conclusion depending on which source appears first.

The athlete-band pattern also matters for interpretation. The aggregate programmatic advantage is driven mainly by beginner and developing profiles, where safe weekly structure may be more constrained: conservative total load, rest-day placement, and hard-day spacing leave less room for stylistic variation. This is an interpretation rather than a demonstrated causal mechanism, but it fits the marker evidence: lower LLM-source rates appear on criteria tied to weekly training structure. For advanced and competitive profiles, LLM-source plans approach parity in the primary forced judgment, suggesting that source preference can depend on the task regime rather than appearing as a single global bias. However, marker-level LLM win rates remain below 50% for every athlete band, so this moderation should not be read as evidence for a general LLM-source advantage.

The order result separates two quantities that are often conflated in LLM-as-judge work. One can estimate an order-balanced aggregate preference by swapping AB/BA positions, but that does not imply that the judge is making a stable pairwise de-

cision. In this study, many pair-judge-run units are position-consistent rather than source-consistent, and the marker pass shows that the apparent source effect changes across evaluation criteria. The practical lesson is that order swaps should not be treated as a cosmetic robustness check; they are necessary for measuring whether a pairwise preference is reproducible under presentation changes.

The study has important limitations. It measures LLM judge preference, not objective coaching quality, and lacks human expert ratings. It uses only Qwen and Gemma judges in one structured generation domain. Structural matching controls observable weekly structure but not all quality-relevant coaching logic or presentation differences. The leakage-filtered analysis is post hoc, and the marker categories are diagnostic labels rather than validated constructs. Finally, the row-level confidence intervals in the appendix should not be read as independent-sample inference because the same matched pairs are judged repeatedly; a stronger follow-up would use pair-clustered bootstrap intervals or a mixed-effects model. These caveats limit claims about training-plan quality, but they strengthen the central evaluation claim: source-preference estimates require order swaps, repeated-judgment diagnostics, criterion decomposition, uncertainty reporting, and checks for presentation leakage.

6 Conclusion

LLM judges in this setting do not prefer directly LLM-source plans over structurally matched programmatic controls: the LLM-source win rate is 36.53% overall. More importantly, the 42.02 percentage-point Plan A swing is larger than the aggregate source effect, and source preference varies by athlete profile and marker. Source-preference studies should therefore treat order swaps, repeated-judgment diagnostics, criterion decomposition, uncertainty reporting, and checks for presentation leakage as required parts of the design.

References

- Jiawei Gu and 1 others. 2024. [A Survey on LLM-as-a-Judge](#).
- Sungeun Kim and Dongsuk Oh. 2025. Evaluating Creativity: Can LLMs Be Good Evaluators in Creative Writing Tasks? *Applied Sciences*, 15(6):2971.
- Yuxuan Liu, Tianchi Yang, Shaohan Huang, Zihan Zhang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, and Qi Zhang. 2024. [HD-Eval: Aligning Large Language Model Evaluators Through Hierarchical Criteria Decomposition](#). *Preprint*, arXiv:2402.15754.
- Kayla Schroeder and Zach Wood-Doughty. 2025. [Can You Trust LLM Judgments? Reliability of LLM-as-a-Judge](#). *Preprint*, arXiv:2412.12509.
- Xiaoyi Tian and 1 others. 2024. Examining LLM Prompting Strategies for Automatic Evaluation of Learner-Created Computational Artifacts. In *Proceedings of the 17th International Conference on Educational Data Mining*, pages 698–706.
- Qiyuan Zhang and 1 others. 2025. Crowd Comparative Reasoning: Unlocking Comprehensive Evaluations for LLM-as-a-Judge. *Advances in Neural Information Processing Systems*.
- Lianmin Zheng and 1 others. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems*, 36:46595–46623.

A Pairwise Judge Prompt Interface

The primary pairwise task was executed through `trailtraining.llm.soft_eval.compare_plans`. The repository does not store a literal natural-language pairwise prompt string; prompt construction occurs inside that external dependency. The repository-controlled and archived pairwise input is the canonical masked JSON payload passed to `compare_plans`, together with the judge configuration. The call used `SoftEvalConfig(enabled=True, reasoning_effort="none", skip_synthesis=True, parallel_batches=False, temperature=0.0)` and returned preferred, reasoning, plan_a_advantages, and plan_b_advantages. Appendix file `appendix_artifacts/pairwise_prompt_interface.txt` records this interface, and `appendix_artifacts/sample_pairwise_judge_input.json` contains a complete archived pairwise payload.

B Marker-Level Judge Prompt

The marker-level evaluation used the following system prompt exactly.

You are an expert running coach and careful evaluator.

You will compare two anonymized training plans, Plan A and Plan B.
The plans have been source-masked. Do not infer or mention whether either plan was generated by a model, a program, or a human.

Evaluate each marker independently. For every marker, choose:

- "plan_a" if Plan A is better on that marker,
- "plan_b" if Plan B is better on that marker,
- "tie" if they are meaningfully equivalent.

Use only the visible plan content. Do not use source labels, file names, IDs, or provenance.
Return valid JSON only. No markdown.

The user prompt template was:

Compare Plan A and Plan B on the following markers.

MARKERS:

- plan_coherence: Overall training-plan coherence: whether the week fits together as a coherent training plan, with compatible sessions, sensible sequencing, and no internal contradictions.
- training_specificity: Specificity to the athlete and trail-running context: terrain, hills, long-run needs, race phase, and appropriate trail/endurance demands.
- load_progression: Appropriateness of training load and progression: total volume, session duration, stress distribution, and whether the plan avoids abrupt or excessive load.
- recovery_safety: Recovery, rest, and safety: rest-day placement, hard-day spacing, injury-risk control, and avoidance of unsafe combinations.
- quality_session_design: Design of quality sessions: workouts such as intervals, tempo, hills, progression runs, or long runs are purposeful, specific, and feasible.
- endurance_development: Support for aerobic/endurance development: easy volume, long-run stimulus, sustainable aerobic work, and endurance-oriented structure.
- readiness_alignment: Alignment with readiness/recovery constraints: whether the plan respects the athlete's readiness, recovery capacity, and fatigue constraints.
- clarity_actionability: Clarity and actionability: whether the athlete can understand and execute the plan without ambiguity.
- explanation_quality: Quality of explanation/rationale: whether the stated purpose and explanation are helpful, specific, and support the plan. This marker is presentation-sensitive.

Scoring guidance:

- Scores are 1 to 5, where 5 is best.
- The preferred plan should normally be the one with the higher score.
- Use tie when differences are negligible.
- Keep rationales short.

REQUIRED JSON SCHEMA:

```
{
  "markers": {
    "plan_coherence": {
      "preferred": "plan_a | plan_b | tie",
      "plan_a_score": "integer 1-5",
      "plan_b_score": "integer 1-5",
      "confidence": "number 0.0-1.0",
      "rationale": "brief reason, max 18 words"
    },
    ...
  },
  "overall_marker_summary": "one short sentence"
}
```

PLAN INPUT:

```
{compact_json(judge_input)}
```

A complete instantiated marker prompt for one archived comparison is included as `appendix_artifacts/sample_complete_marker_prompt.txt`.

C Canonical Judge-Facing Schema

Both sources are rendered into the same canonical judge-facing schema. The judge sees only neutral `plan_a` and `plan_b` objects.

```
{
  "plan_a": {
    "judge_view": "canonical_masked_structural_v1",
    "plan": {
      "days": [
        {
          "day_index": "integer 1..7",
          "session_type": "string",
          "duration_minutes": "integer",
          "is_rest_day": "boolean",
          "is_hard_day": "boolean",
          "target_intensity": "string",
          "workout_summary": "short normalized text",
          "purpose_summary": "short normalized text"
        }
      ],
      "summary": {
        "plan_days": "integer",
        "total_minutes": "integer",
        "rest_days": "integer",
        "hard_days": "integer",
        "long_runs": "integer"
      }
    }
  },
  "plan_b": "same schema as plan_a"
}
```

The source family, judge family, source model, plan IDs, structural-score gap, self-family flag, AB/BA order, judge identity, and run index are retained in the manifest outside the prompt for analysis. The frozen artifact set is named `canonical_masked_scrubbed_v1_t000`; the judge-facing payload inside each archived JSON record uses the `canonical_masked_structural_v1` view shown above.

D Structural-Score Feature Definitions

The structural score excludes title wording, prose richness, citations, claim attributions, data-note verbosity, rationale or explanation fields, source model, generation arm, and file names. It uses only plan structure and fixture-conditioned expectations. The scored features are:

- **plan days:** actual number of plan days compared with expected length;
- **total minutes:** sum of daily `duration_minutes`;
- **active, rest, hard, quality days:** counts derived from rest and hard flags and quality session types;
- **long runs:** count of `session_type == long`;
- **maximum and mean day minutes:** daily load distribution summaries;
- **hard-day spacing:** minimum distance between hard days;
- **fixture-adjusted load, hard-day, and rest-day expectations:** ranges conditioned on athlete band, readiness, recovery capability, and race phase.

The matching distance uses weighted gaps in total minutes, number of rest days, number of hard days, number of active days, number of long runs, number of quality days, maximum day minutes, and mean day minutes. Soft calipers penalize large gaps in total minutes, maximum day minutes, rest days, hard days, long runs, and quality days. Hard calipers were not enabled in the frozen primary match because the target-cardinality design prioritized retaining 250 same-cell pairs. The machine-readable definition is included as `appendix_artifacts/structural_score_features.json`.

E Sample Judge-Facing JSON

The following excerpt shows one canonical comparison. The full JSON appears in `appendix_artifacts/sample_pairwise_judge_input.json`.

```
{
  "plan_a": {
    "judge_view": "canonical_masked_structural_v1",
    "plan": {
      "days": [
        {
          "day_index": 1,
          "duration_minutes": 48,
          "is_hard_day": false,
          "is_rest_day": false,
          "purpose_summary": "Build aerobic consistency while keeping fatigue low.",
          "session_type": "easy",
          "target_intensity": "easy",
          "workout_summary": "48 min easy run at conversational effort."
        },
        {
          "day_index": 2,
          "duration_minutes": 39,
          "is_hard_day": false,
          "is_rest_day": false,
          "purpose_summary": "Build aerobic consistency while keeping fatigue low.",
          "session_type": "easy",
          "target_intensity": "easy",
          "workout_summary": "39 min easy run at conversational effort."
        }
      ],
      "summary": {
        "hard_days": 0,
        "long_runs": 0,
        "plan_days": 7,
        "rest_days": 2,
        "total_minutes": 229
      }
    }
  },
  "plan_b": {
    "judge_view": "canonical_masked_structural_v1",
    "plan": {
      "days": [
        {
          "day_index": 1,
          "duration_minutes": 38,
          "is_hard_day": false,
          "is_rest_day": false,
          "purpose_summary": "Build confidence and endurance.",
          "session_type": "easy",
          "target_intensity": "easy",
          "workout_summary": "Easy trail run focusing on form and cadence."
        },
        {
          "day_index": 2,
          "duration_minutes": 41,
          "is_hard_day": false,
          "is_rest_day": false,
          "purpose_summary": "Maintain aerobic base and hill strength.",
          "session_type": "easy",

```



```

        "target_intensity": "easy",
        "workout_summary": "Easy trail run to maintain aerobic base."
    }
],
"summary": {
    "hard_days": 0,
    "long_runs": 0,
    "plan_days": 7,
    "rest_days": 1,
    "total_minutes": 321
}
}
}
}

```

F Sample Metadata Retained Outside the Prompt

The manifest retains analysis metadata that the judge does not see. One example is shown below; the full file appears as `appendix_artifacts/sample_pair_metadata_outside_prompt.json`.

```

{
  "record_id": "pair_0001__qwen_14b_judge__r00__BA",
  "pair_id": "pair_0001",
  "fixture_id": "ab_A1__r_high__rc_high__ph_base",
  "judge": "qwen_14b_judge",
  "judge_model_family": "qwen",
  "run": 0,
  "order": "BA",
  "left_plan_role": "programmatic",
  "right_plan_role": "llm",
  "llm_plan_id": "ab_A1__r_high__rc_high__ph_base__qwen2.5_7b_instruct__src_t070__exp_t000__s004",
  "programmatic_plan_id": "ab_A1__r_high__rc_high__ph_base__prog__exp_t000__s008",
  "source_model_family": "qwen",
  "self_family_match": true,
  "structural_score_gap": 0.0,
  "structural_score_version": "structural_score_v1.0.0",
  "feature_gaps": {
    "max_day_minutes": 35.0,
    "mean_day_minutes": 13.143,
    "n_active_days": 1.0,
    "n_hard_days": 0.0,
    "n_long_runs": 0.0,
    "n_quality_days": 0.0,
    "n_rest_days": 1.0,
    "total_minutes": 92.0
  },
  "judge_view": "canonical_masked"
}

```

G Detailed Results Tables

Winner	Count	Rate
LLM-source	3,653	36.53%
Programmatic	6,347	63.47%

Table 4: Primary forced pairwise outcome.

H Sample Judge-Facing JSON and Metadata

Full examples are archived in `appendix_artifacts/sample_pairwise_judge_input.json` and `appendix_artifacts/sample_pair_metadata_outside_prompt.json`. The main report omits the full JSON listing to keep the page-limited portion focused on the experimental design and results.

Band	Profile	Pairs	Primary	Marker
A1	Beginner	60	12.50%	27.16%
A2	Developing	55	24.55%	28.89%
A3	Advanced	74	51.08%	44.45%
A4	Competitive	61	53.32%	45.96%

Table 5: LLM-source win rates by athlete band. Primary is the forced pairwise win rate. Marker is the LLM-source win rate excluding ties across all marker decisions.

Judge	LLM as Plan A	LLM as Plan B	Plan-A advantage
Gemma 12B	39.60%	19.60%	20.00 pp
Gemma 4B	62.00%	4.88%	57.12 pp
Qwen 14B	74.40%	8.40%	66.00 pp
Qwen 7B	54.16%	29.20%	24.96 pp
Overall	57.54%	15.52%	42.02 pp

Table 6: Order effect by judge. Rates are LLM-source win rates when the LLM-source plan appears as Plan A versus Plan B.

Pair-run category	Count	Rate
Source-consistent programmatic	1,973	39.46%
Source-consistent LLM-source	626	12.52%
Position-consistent Plan A	2,251	45.02%
Position-consistent Plan B	150	3.00%

Table 7: AB/BA pair-run consistency over 5,000 matched pair-judge-run units.

Marker	LLM wins	Prog. wins	Ties	LLM win rate excl. ties
Training specificity	4,368	4,630	1,002	48.54%
Clarity/actionability	2,710	3,962	3,328	40.62%
Readiness alignment	2,863	4,324	2,813	39.84%
Recovery safety	3,103	4,839	2,058	39.07%
Plan coherence	2,835	5,040	2,125	36.00%
Load progression	3,218	5,947	835	35.11%
Quality session design	2,783	5,539	1,678	33.44%
Endurance development	3,046	6,239	715	32.81%
Explanation quality	2,578	5,425	1,997	32.21%

Table 8: Marker-level outcomes across 10,000 records per marker. The final column excludes ties; tie counts are shown because the marker task permits equivalence judgments.

Comparison	<i>n</i>	LLM wins	Rate	CI low	CI high
Overall	10,000	3,653	36.53%	35.59%	37.47%
LLM as Plan A	5,000	2,877	57.54%	56.17%	58.91%
LLM as Plan B	5,000	776	15.52%	14.52%	16.52%

Table 9: Descriptive row-level Wald 95% confidence intervals. These are not independent-sample inference because rows repeat matched pairs across judges, runs, and orders.

Sample	Pairs	Judgments	LLM wins	Prog. wins	LLM win rate
Full sample	250	10,000	3,653	6,347	36.53%
Artifact-clean subset	103	4,120	1,419	2,701	34.44%
Artifact-present subset	147	5,880	2,234	3,646	37.99%

Table 10: Full-sample and leakage-filtered sensitivity analysis. The artifact-clean subset excludes any matched pair flagged by the post hoc presentation-artifact audit.